RV College of Engineering®, Bengaluru – 59
Department of Artificial Intelligence and Machine
Database Management Systems (CD252IA)

Synopsis

TITLE	**Intelligent Inventory Management System using Data-Driven Forecasting**	
TEAM	1RV23AI135	Abhinav Potharaju
	1RV23AI132	Vineet Raj

1. Introduction

This project focuses on building an intelligent, data-driven forecasting system for the retail sector using historical sales data. It applies advanced analytics and machine learning to predict demand accurately, helping businesses optimize inventory, reduce costs, and improve supply chain efficiency.

2. Existing System

Many small and medium-sized enterprises (SMEs) in the retail sector still rely on traditional inventory management methods, such as manual tracking through spreadsheets. These practices are inefficient, error-prone, and lack scalability for growing business needs. Basic statistical models like moving averages remain reactive, adjusting only to past trends instead of predicting future demand. Moreover, data silos separating transactional data from customer feedback limit analytical depth, reducing forecast accuracy and causing issues like overstocking and stockouts.

3. Proposed System

The proposed system is an intelligent, data-driven forecasting model that uses historical sales data and machine learning to predict demand accurately. It helps businesses optimize inventory, reduce operational costs, and improve supply chain efficiency and responsiveness.

4. Relational Database Structure

- **Database:** Implemented in **MySQL** as the main repository for all structured data.
- **Normalization:** Designed up to **Third Normal Form (3NF)** to ensure data integrity, consistency, and eliminate redundancy.
- **Core Tables:**
    - **Customers:** CustomerID (PK), Country
    - **Products:** StockCode (PK), Description
    - **Invoices:** InvoiceNo (PK), InvoiceDate, CustomerID (FK)
    - **Invoice_Items:** InvoiceNo (FK), StockCode (FK), Quantity, Price
- **Schema Extensions:**
    - **Holidays:** HolidayDate (PK), HolidayName

- **Promotions:** PromotionID (PK), StockCode (FK), StartDate, EndDate

5. RDBMS and NoSQL Integration

- **Architecture Strategy:** Uses a **polyglot persistence** approach, combining **relational (MySQL)** and **NoSQL (MongoDB)** databases to manage different data types and workloads efficiently.
- **MySQL (RDBMS) – System of Record:**
    - Stores **structured transactional data** such as sales, customers, holidays, and promotions.
    - Ensures **data integrity, accuracy, and consistency** using primary/foreign keys and constraints.
    - Follows **ACID properties** for reliable transactions and serves as the **official business ledger**.
- **MongoDB (NoSQL) – System of Engagement:**
    - Handles **semi-structured and unstructured data**, including product catalogs and customer reviews.
    - Offers **flexibility and scalability**, adapting to frequently changing or varied data formats.
- **Integration Process:**
    - A **Python-based ETL pipeline** integrates MySQL and MongoDB data at the application layer.
    - Structured data from MySQL is enriched with contextual information from MongoDB to create a **comprehensive dataset** for **AI-driven, context-aware forecasting**.

6. Societal Concern

- **Societal Impact:** The project tackles the challenge of **Supply Chain Optimization and Sustainability** by reducing waste and resource misuse caused by poor inventory management.
- **Goal:** Accurate, data-driven forecasting aligns inventory with real consumer demand, minimizing overproduction, cutting costs, and lowering environmental impact.
- **Technological Integration:**
    - **Predictive Forecasting (ML):** Uses historical and enriched data to produce precise demand forecasts.
    - **Personalization Engine:** Applies **Market Basket Analysis (Apriori Algorithm)** to find product associations and enable smart recommendations.
    - **Natural Language Processing (NLP):** Analyzes customer reviews for **sentiment scoring** and uses an **LLM** to summarize forecast insights for non-technical users.
    - **Real-Time Demand Sensing:** Dynamically updates forecasts based on events, promotions, or market trends.
    - **Full-Stack Architecture:** Built with a **React frontend** and **Flask backend API**, providing scalability, maintainability, and easy accessibility.